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# EarStudio: DualDrive Technology

Dual DAC/AMP scheme for Single-Ended and Balanced Outputs

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Due to its outstanding performance of the full differential signal path, full-differential (a.k.a. balanced) output has become an essential feature of hi-fi audio players or amplifiers. As shown in Fig.1, more and more headphone amplifiers and portable devices started to add differential output.



**Fig.1. Audio devices with Full-Differential output**

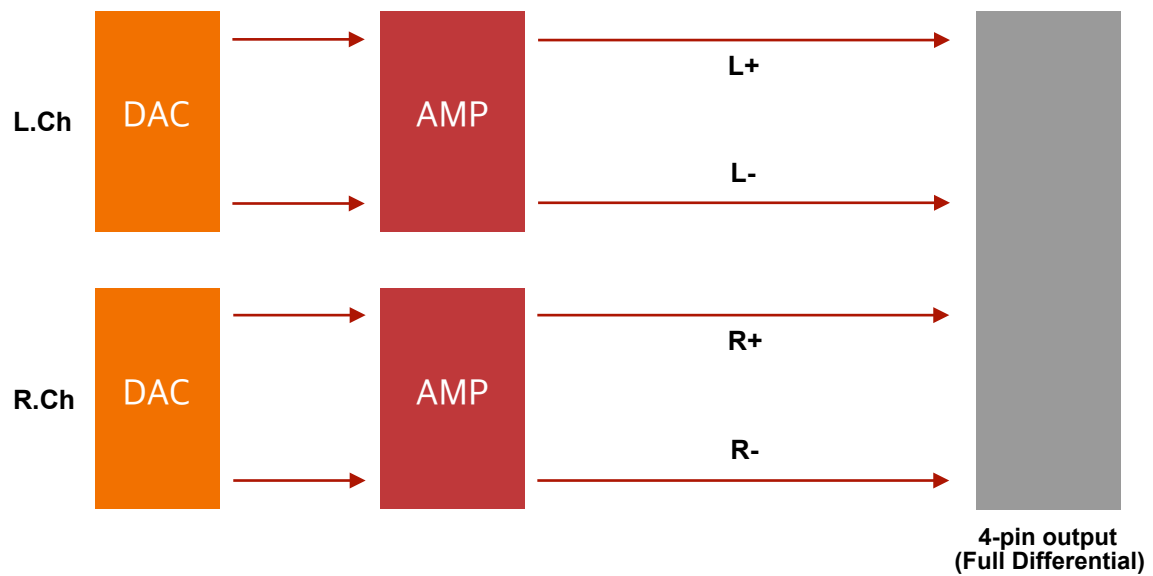
Since devices with the balanced output have begun to catch audiophile's attention, not only flagship models but mid-end models have started to support the detachable cable interface for the full-differential connection with the devices, escalating the necessity of the full-differential interface.



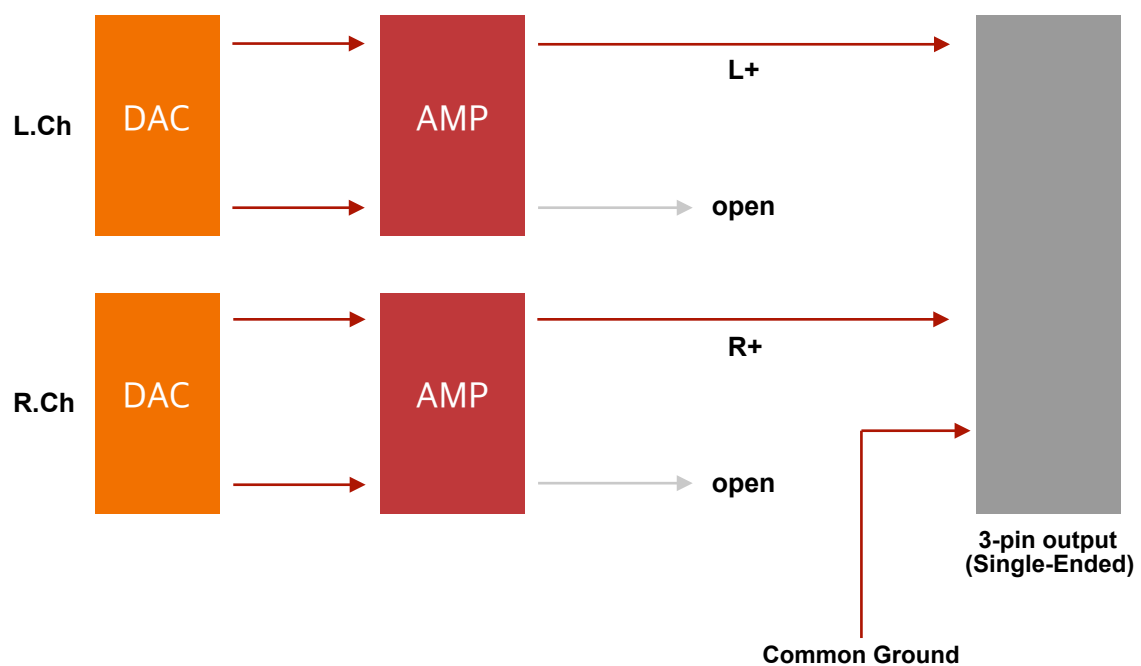
**Fig.2. Headphones/Earphones with detachable cable interface**

As single-ended (a.k.a Unbalanced) 3-pin (L, R, and GND) cable is most prevalent in the market with non-detachable cable, the audio devices require supporting both of the standard single-ended interface as well as the full-differential interface.

Supporting both S/E (Single-Ended) and F/D (Full-Differential) output modes in a single audio device makes the system design complicated. Mostly it requires switching element to disconnect and isolate the output signals from the unused output port.

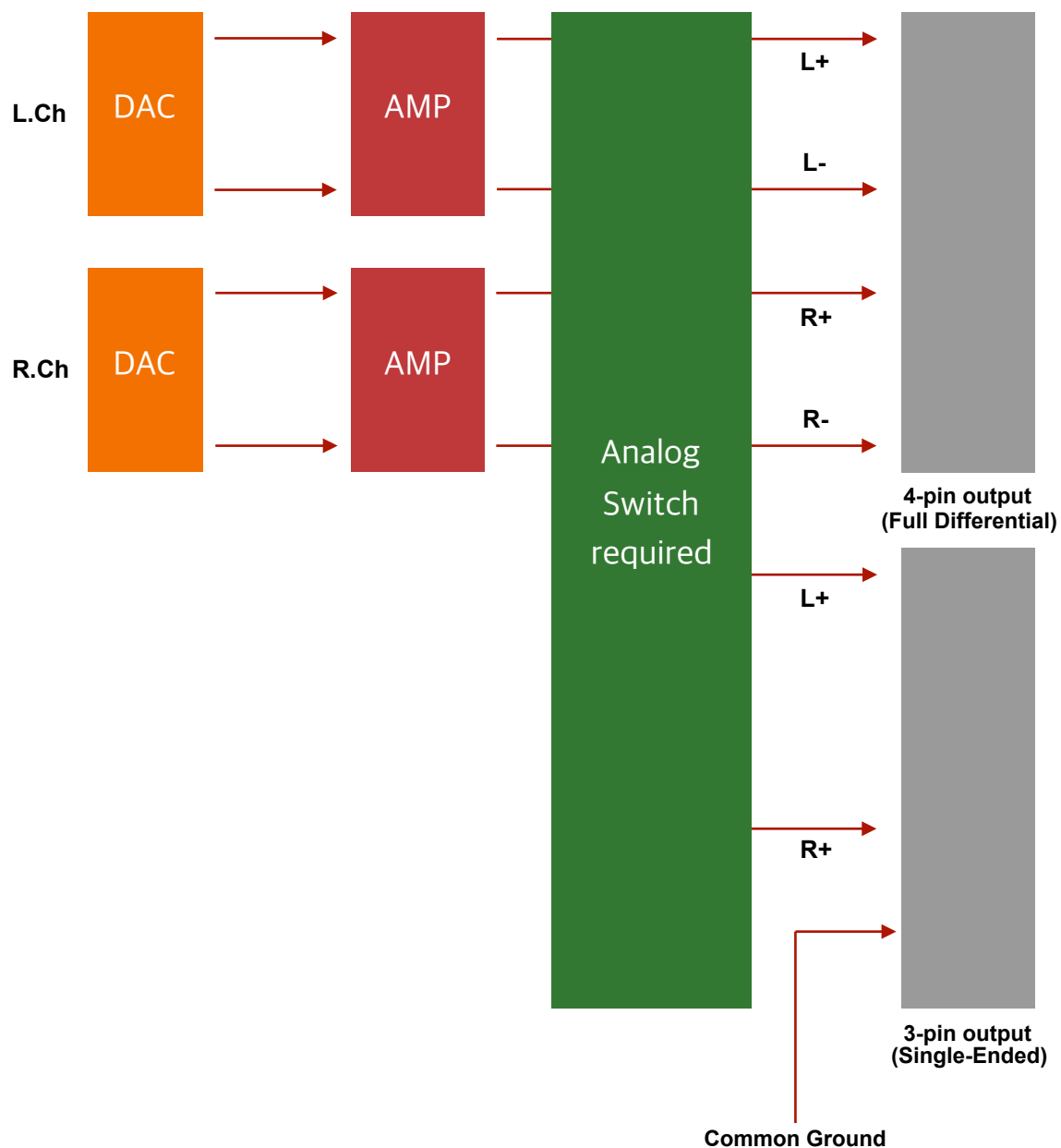


**a) Full-Differential 4-pin output**



**b) Single-Ended 3-pin output**

**Fig.3. Full-Differential (F/D) vs. Single-Ended (S/E)**



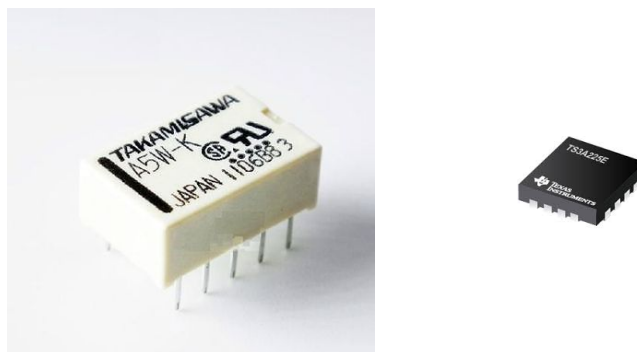
**Fig.4. Analog switch for F/D and S/E output**

Fig.4 shows an example design of a conventional audio device supporting both the S/E and the F/D output ports. As shown in the figures, analog switch components are required for the exclusive output on/off control of each port.

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There are two kinds of analog switches: Relay and Semiconductor switch

The relay switch is mechanically operated and provides minimal loss through its signal path. But, it's slow and consumes additional power to hold the switch with its built-in electromagnet. Moreover, the size is too big to be integrated into portable devices.

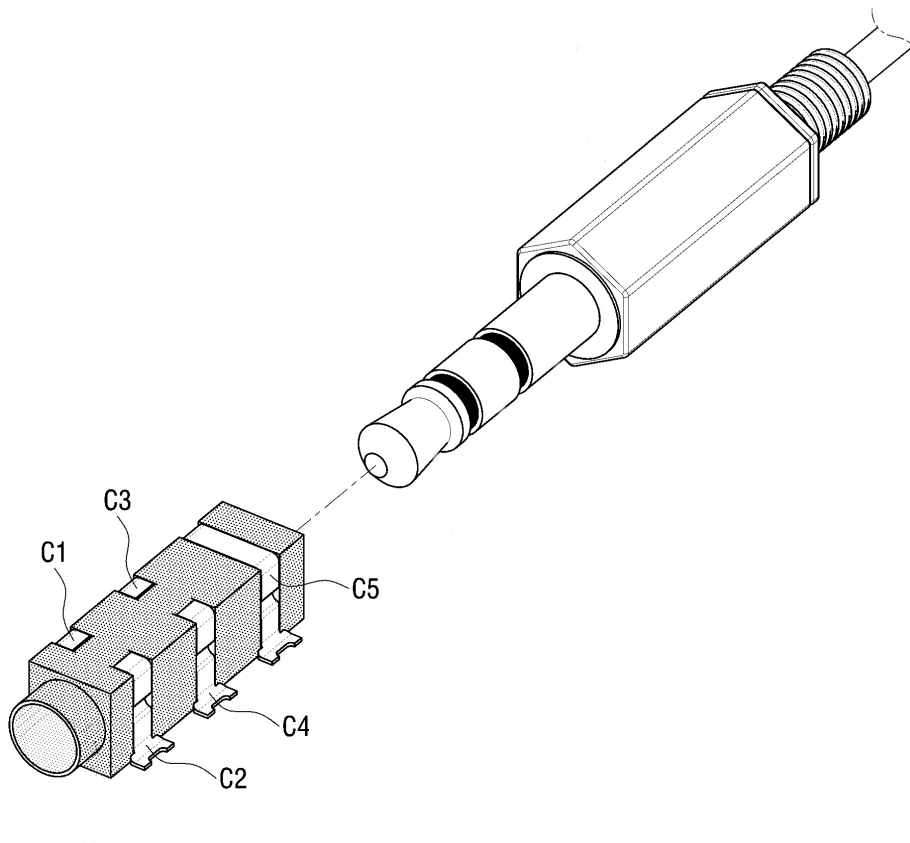


**Fig.5. Analog switch: Relay vs. Semiconductor**

On the other hands, semiconductor switch (a.k.a. solid-state switch) is fast, small, easy-to-control, and power efficient. However, there is significant loss through its path, which eventually degrades the given output performance of the amplifier.

When the semiconductor switch is on, there is always some resistance due to its nature. For delivering the digital/logic signal, those resistances isn't so critical to the performance and the functionality. But for the analog signal, the On-State Resistance of semiconductor switch (a.k.a  $R_{on}$ ) contributes to signal loss and degradation. Also, the switch on/off capacitance must be considered carefully especially for the high-end audio application.

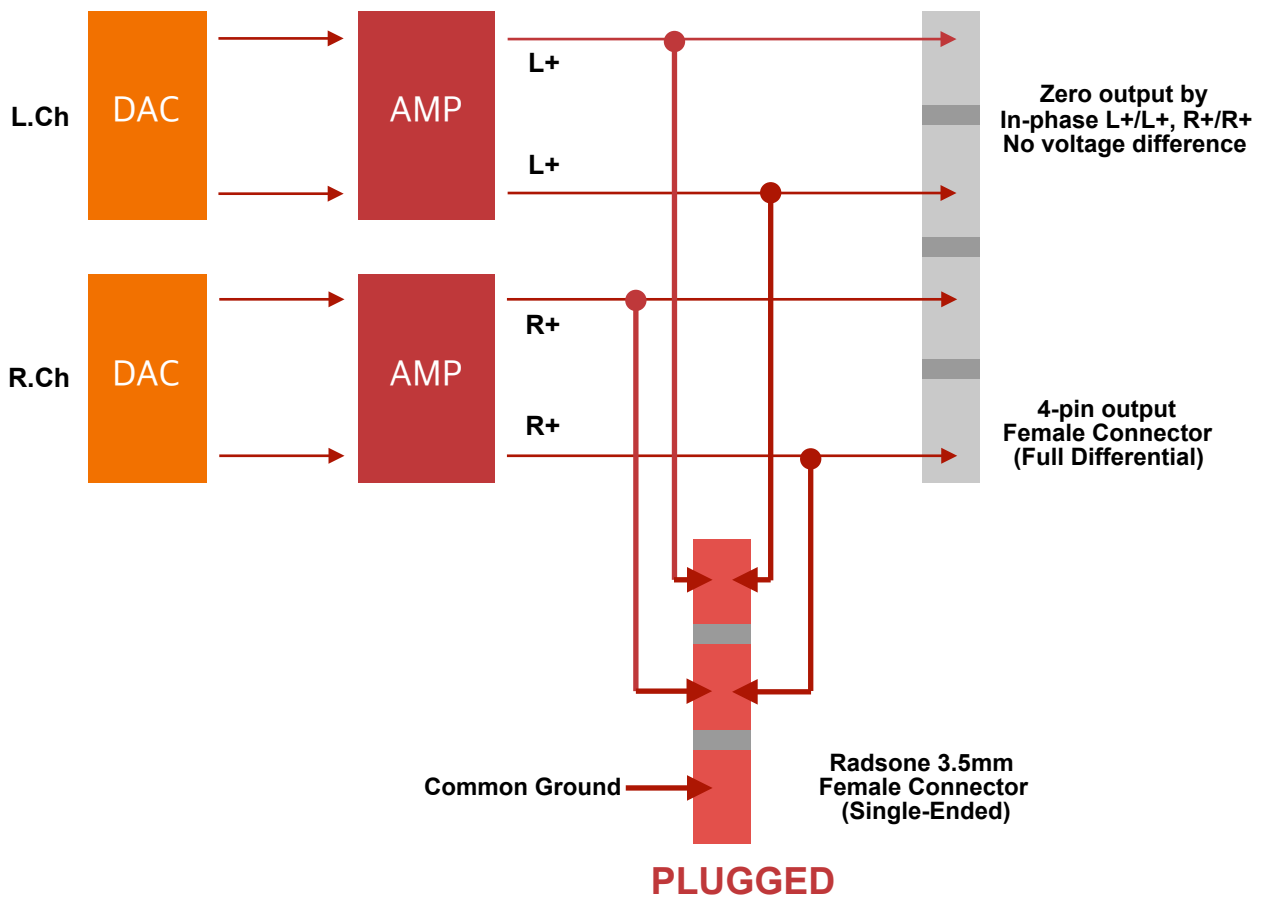
For those reasons, Radsone invented and introduced DualDrive architecture, providing the simplest and the best way of designing the system without any switching elements.



**Fig.6. Radsone's single-ended female connector for DualDrive**

Fig.6 illustrates a single-ended female audio connector designed by Radsone for DualDrive technology.

Unlike other conventional 3.5mm female connectors, C1 and C2, C3 and C4 in Fig.3 are normally open when unplugged. When plugged, the 3.5mm male connector electrically interlinks C1 and C2, C3 and C4 respectively, operating as a mechanical switch. It is the principal idea of the Radsone's 3.5mm female connector, contributing the primary role to DualDrive technology as the main component.



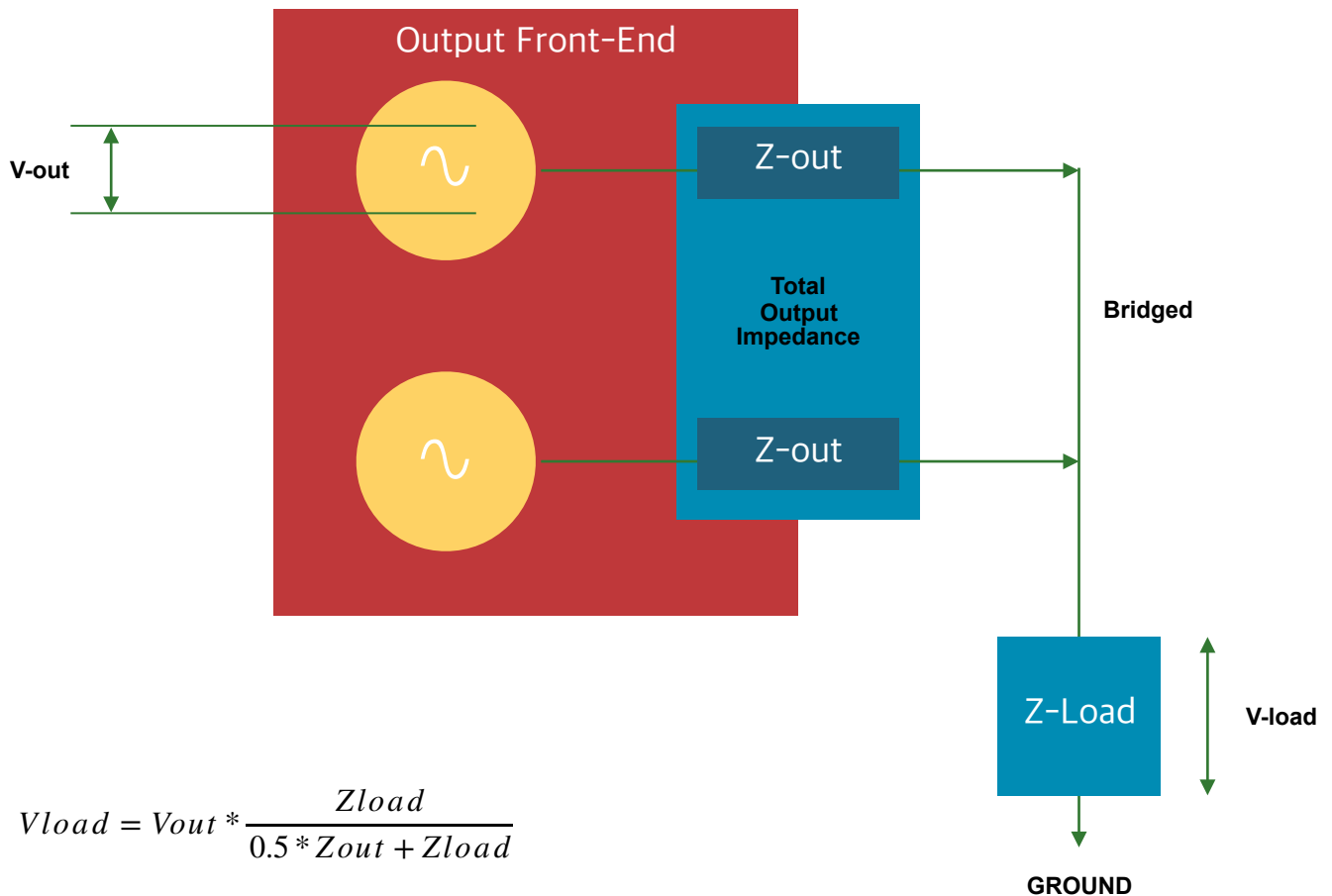
**Fig.7. DualDrive (Single-Ended output) - 2x Current Mode**

As described in Fig.7, DualDrive technology has NONE of switching element in the way from the output stage to the female output connector. Radstone's single-ended female connector works as a switch which gets on by the connected male audio jack.

When the standard 3.5mm audio jack is plugged into Radstone's 3.5mm female connector, the system gets operated in a bridged mode. The each DAC is configured having both analog outputs to be in-phase and identical: (L+/L+, R+/R+). The connected male audio jack couples and combines those +/+ output signals together by itself.

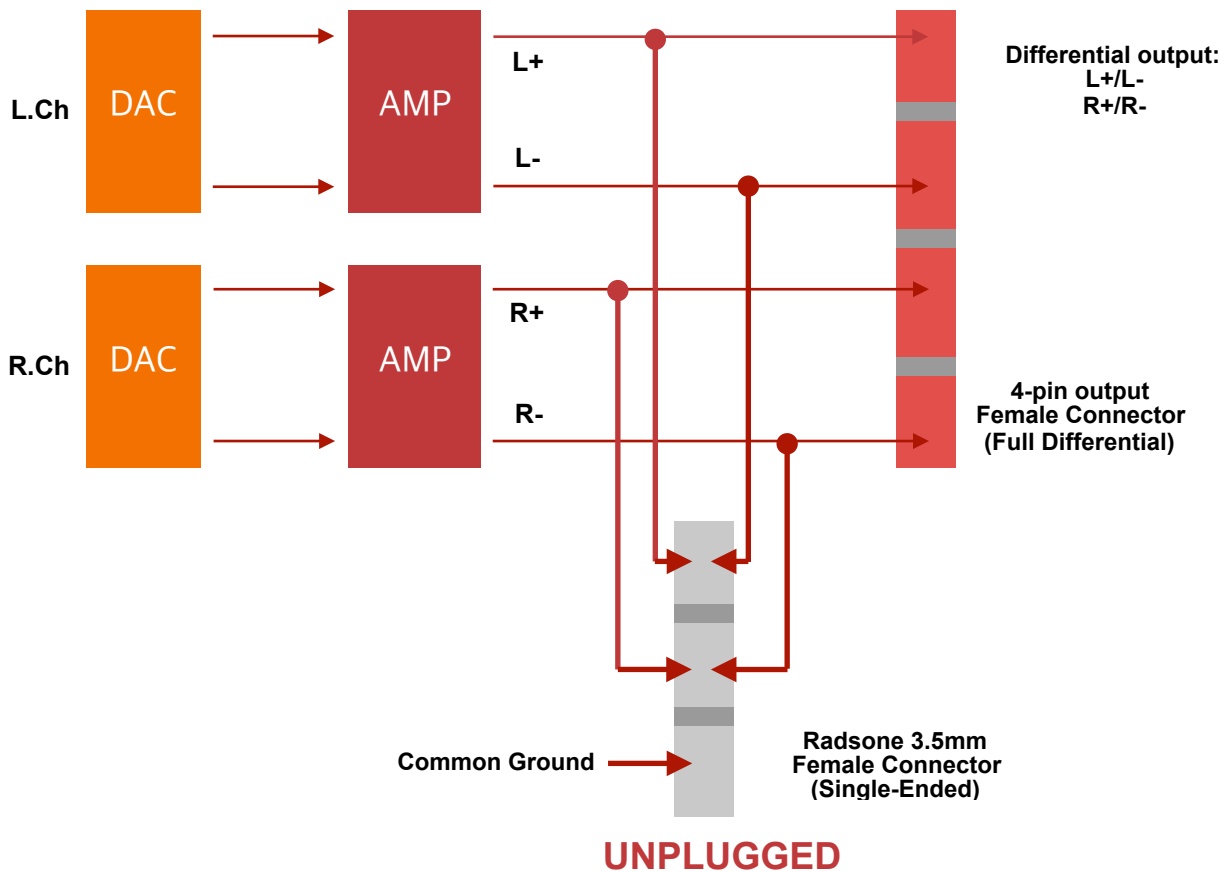


Every output signal from each amplifier still goes to the Full-Differential output port directly, but because there is no difference in voltage level between L+ and L- pin, or R+ and R- pin, NO output will be delivered to the Full-Differential port.



**Fig.8. Output Impedance of Bridged Mode**

In that case, the device can deliver twice much current. Also, the output impedance of the given system gets lowered by half, as described in Fig.8. As result of which, the damping factor and slew rate are increased, especially for those low impedance headphone units.



**Fig.9. DualDrive (Full-Differential output) - 2x Voltage Mode**

When the system detects 3.5mm standard connector unplugged, it changes to the full-differential operating mode by inverting the lower output signals from positive to negative. Eventually, the L+ and L-, R+ and R- are delivered to the output connector directly; the same way of any balanced audio system, but NO switches.

In conclusion, our proprietary DualDrive architecture provides the best and the optimized way of designing Dual DAC/AMP system for both single-ended (unbalanced) and full-differential (balanced) outputs in a single device. No switching parts are required with DualDrive architecture, enabling smaller PCB footprint, BOM cost reduction, and the better audio performances.